7 The radioactive isotope of Bismuth $^{210}_{83}Bi$ decays into Polonium Po with the emission of a beta particle.

(a) **(i)** Write down the equation representing the beta decay of $^{210}_{83}Bi$.

.....[2]

(ii) State two quantities that are conserved in any radioactive decay process.

.....

.....[2]

(b) **(i)** The mass of a $^{210}_{83}Bi$ nucleus is 209.939u.
Show that its mass defect is 1.767u.
(mass of proton, m_p = 1.00729u; mass of neutron, m_n = 1.00867u)

[2]

(ii) Calculate the binding energy per nucleon, in MeV, of $^{210}_{83}Bi$. Leave your answer to 3 significant figures.

binding energy per nucleon = ... MeV [2]

[Total: 8]